Supplementary Material S1

Search Strings by Platform

Facial Emotion Recognition through Machine Learning in Children with Autism Spectrum Disorder: A systematic scoping review of open-access computational evidence and of the gaps toward therapeutic monitoring

1 Summary of execution

The three main searches (Scopus, Web of Science Core Collection and IEEE Xplore) and the complementary search in ACM Digital Library were executed on 15 February 2026. The four strings share the same Boolean structure and descriptors, organised into three conceptual blocks joined by AND: population (ASD), concept (facial emotional expressiveness) and method (machine learning). The differences across platforms are field syntax and temporal restriction. The three main searches were updated on 12 September 2026; the update did not identify additional eligible studies.

Platform	**Temporal range**	**Records retrieved**	**Execution date**
Scopus	2019–2025	935	15/02/2026
Web of Science Core	2018–2026	942	15/02/2026
IEEE Xplore	2018–2026	38	15/02/2026
ACM Digital Library	2018–2026	5	15/02/2026
Total		**1,920**	

Distribution by platform at each stage of the selection process:

Stage	**Scopus**	**WoS**	**IEEE**	**ACM**	**Total**
Identification (records retrieved)	935	942	38	5	1,920
Duplicates removed	0	504	20	0	524
Screening (unique records)	935	438	18	5	1,396
Full-text assessment	62	24	8	5	99
Eligible corpus	24	4	2	0	30
Contextual evidence	9	5	1	0	15

Duplicates were removed with `pipeline/deduplicate_records.py`: exact match on the normalised DOI first, then on the normalised title and publication year, with precedence Scopus, Web of Science, IEEE Xplore and ACM Digital Library. Each duplicate is assigned to the platform whose record was discarded. All 524 duplicates matched by DOI; none required the title and year rule. The resulting list of 1,920 records with their duplicate flags is `data/records_deduplicated.csv`, and the decision, criterion and reason for each of the 1,396 unique records is `data/screening_log.csv`.

The version of this document before V3.6 reported 1,007 records removed before screening and 913 records screened. The reconstruction showed that 479 unique records had been removed without a documented reason; they were screened in V3.6 and are included in the counts above. The audit of criterion CI(3) against the SCImago editions of each publication year (`data/sjr_verification.csv`) moved five studies without an SJR quartile in that year from

the eligible corpus to contextual evidence, and the full-text check of 12 November 2026 moved Cicceri et al. (2026) to contextual evidence under CE(3). The five ACM records had been excluded before screening for lack of open access (CI(4)). This was not valid, because the ACM Digital Library has been fully open access since 1 January 2026, before the search date; in V3.6 they were assessed in full text by content (three fail CI(1) and two fail CI(3); `data/ACM-data.csv`).

Of the 99 reports assessed in full text, 30 constitute the eligible corpus, 15 are documented as contextual evidence outside the eligible set, 8 were retracted or withdrawn, and 46 were excluded with a recorded criterion (PRISMA 2020 flow diagram in `prisma/PRISMA_2020.jpg`; full list in Supplementary Material S5).

Forward and backward citation chasing identified five candidate studies; their full-text screening was completed during the September 2026 update and none met the eligibility criteria, so this procedure did not modify the eligible corpus.

The Scopus string was restricted to 2019–2025, whereas the Web of Science and IEEE Xplore strings applied 2018–2026. The aggregate coverage of the three main searches spans the full 2018–2026 conceptual window: Web of Science contributed the single eligible study published in 2018 and IEEE Xplore the single one published in 2026, and the 24 eligible studies retrieved in Scopus fall within 2019–2025. The year 2023 appears with zero eligible studies; the column is retained in the corresponding table of the manuscript to preserve the continuity of the temporal series.

2 Conceptual blocks common to the four platforms

Block	**Descriptors joined by OR**
Population	autism; ASD; autism spectrum; autistic
Concept	emotion; facial expression; affective; social communication
Method	machine learning; deep learning; artificial intelligence; computer vision; neural network; classification; recognition

The three blocks are combined with AND. Filters applied across all four platforms: document type restricted to journal article and conference paper; open access; and English language.

3 String executed in Scopus

Temporal range: 2019–2025. Records retrieved: 935.

```
TITLE-ABS-KEY(("autism" OR "ASD" OR "autism spectrum" OR "autistic") AND ("emotion" OR
"facial expression" OR "affective" OR "social communication") AND ("machine learning"
OR "deep learning" OR "artificial intelligence" OR "computer vision" OR "neural
network" OR "classification" OR "recognition")) AND PUBYEAR > 2018 AND PUBYEAR < 2026
AND (LIMIT-TO(DOCTYPE,"ar") OR LIMIT-TO(DOCTYPE,"cp")) AND (LIMIT-TO(OA,"all")) AND
(LIMIT-TO(LANGUAGE,"English"))
```

TITLE-ABS-KEY searches title, abstract and keywords simultaneously. The filter `PUBYEAR > 2018 AND PUBYEAR < 2026` is exclusive at both ends and therefore covers 2019–2025, as confirmed by the year distribution of the 935 exported records (`data/scopus-data.csv`).

4 String executed in Web of Science Core Collection

Temporal range: 2018–2026. Records retrieved: 942.

```
TS=(("autism" OR "ASD" OR "autism spectrum" OR "autistic") AND ("emotion" OR "facial
expression" OR "affective" OR "social communication") AND ("machine learning" OR "deep
learning" OR "artificial intelligence" OR "computer vision" OR "neural network" OR
"classification" OR "recognition")) AND PY=(2018-2026) AND DT=(Article OR Proceedings
Paper) AND LA=(English)
```

TS= (Topic) is the functional equivalent of TITLE-ABS-KEY in Scopus. PY= restricts by year, DT= by document type and LA= by language. Open access was applied as interface refinement.

5 String executed in IEEE Xplore Digital Library

Temporal range: 2018–2026. Records retrieved: 38.

```
(("All Metadata":"autism" OR "All Metadata":"ASD" OR "All Metadata":"autism spectrum"
OR "All Metadata":"autistic") AND ("All Metadata":"emotion" OR "All Metadata":"facial
expression" OR "All Metadata":"affective" OR "All Metadata":"social communication")
AND ("All Metadata":"machine learning" OR "All Metadata":"deep learning" OR "All
Metadata":"artificial intelligence" OR "All Metadata":"computer vision" OR "All
Metadata":"neural network" OR "All Metadata":"classification" OR "All
Metadata":"recognition"))
```

Executed through Command Search. Temporal restriction (2018–2026), document type (Journals and Conferences), open access and English language were applied as interface refinements. All Metadata is broader than TITLE-ABS-KEY and additionally includes IEEE indexing terms.

6 String executed in ACM Digital Library

Temporal range: 2018–2026. Records retrieved: 5.

```
[[All: "autism"] OR [All: "ASD"] OR [All: "autism spectrum"] OR [All: "autistic"]]
AND [[All: "emotion"] OR [All: "facial expression"] OR [All: "affective"] OR [All:
"social communication"]] AND [[All: "machine learning"] OR [All: "deep learning"] OR
[All: "artificial intelligence"] OR [All: "computer vision"] OR [All: "neural
network"] OR [All: "classification"] OR [All: "recognition"]]
```

Executed through ACM DL Advanced Search. Temporal restriction (01/2018–12/2026), open access and English language were applied as interface refinements. The [All: ”term”] operator searches across all indexed metadata fields. The 5 retrieved records were assessed by content in V3.6; none met the eligibility criteria (three CI(1), two CI(3)).

7 Pilot searches

Pilot searches with therapeutic and gamification terminology were executed to verify coverage:

- serious game AND autism AND emotion recognition
- therapeutic monitoring AND ASD AND machine learning
- mHealth AND autism AND facial expression

These jointly retrieved fewer than 2% of additional non-overlapping records, confirming that the

main string adequately captures the therapeutic monitoring literature in these databases. The limitation is documented in Section 2.9 of the manuscript.

8 Complementary procedures

- Forward and backward citation chasing: five candidate studies identified. Full-text screening completed during the September 2026 update; none of the five met the eligibility criteria in full.

Procedures not executed: formal PRESS review by external specialist librarian; platform similarity recommendation functions; systematic exploration of grey literature.

Search update, 12 September 2026. The three main searches were re-executed on 12 September 2026 with the same strings and eligibility criteria. The citation chasing candidates were also screened to completion at this date. Neither the database update nor citation chasing yielded additional eligible studies.

Screening reconstruction, V3.6 (November 2026). The screening was reconstructed from the exported records as described in Section 1. The former exclusion criterion of fewer than 10 participants was removed, and the scope criterion CI(1) was made explicit: an eligible study must use or evaluate an automated analysis of the face or gaze. After the audit of criterion CI(3) described in Section 1, the eligible corpus comprises N = 31 studies (Supplementary Material S5).